SH E.656

"Caimano"

SH E.656 Polyphase Electric Locomotive

Ver. Classic 1.2



SILVER HIVE
v3 Classic

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1. Introduction

This manual describes the operation of the **SH E.656 «Caimano» Ver. Classic**, an add-on for Train Simulator Classic developed by **Silver Hive (SH)**. The Classic edition provides the essential controls for everyday driving: battery, pantographs, braking system, lights and the integrated Silver Hive RSC/SCMT system.

> ■ **Classic Edition**
> Simplified procedure without PAC, SHUNT or advanced IR protections. The RSC/SCMT system is natively integrated by Silver Hive — no external plugin required.

2. Locomotive Overview

2.1 Real Prototype

The **SH E.656 «Caimano»** reproduces the Italian electric locomotive E.656, built in 461 units between 1975 and 1989. Powered by **3000 V DC**, it was designed for mixed passenger and freight services.

2.2 Technical Specifications

Parameter	Value
Type	Electric locomotive
Power supply	3000 V DC
Continuous power	3480 kW (4 motors × 870 kW)
Max speed	160 km/h
Service mass	~83 t
Wheel arrangement	Bo'Bo'Bo'
Built	1975–1989 | 461 units

2.3 Provider Activation in TSC

Before starting a scenario, enable the Silver Hive provider in the TSC editor.

Step 1 — Select SilverHive provider

In the TSC editor click the **blue square** (circled in red) at the top left. From the dropdown select **SilverHive**: **SH_E656_Classic** will appear in the list.



Figure P1 — Selecting the SilverHive provider

Step 2 — Confirm selection

Verify that **SH_E656_Classic** is selected in the left panel and confirm placement in the scenario.



Figure P2 — SH_E656_Classic selected and ready

Half-bodies A and B

The locomotive is made of two separate half-bodies, **A** and **B**, each with its own cab. In the editor they appear as separate entries (**SH_E656_A_Classic** and **SH_E656_B_Classic**): to obtain the complete locomotive, insert both, coupled as shown.



Figure P3 — Half-body A and half-body B in the TSC editor

XMPR and TEE variants

The **XMPR** and **TEE** variants are available for both half-bodies (for example **SH_E656_A_Classic_XMPR**, **SH_E656_B_Classic_TEE**).



Figure P4 — XMPR variant



Figure P5 — TEE variant

Double traction and multiple-unit control

When two locomotives are coupled in line they are recognised automatically: double traction is active with no further operation. In the second locomotive the driver is not present.



Figure P6 — Double traction: locomotives coupled in line

> ■ **No external plugin required**
> The RSC/SCMT system is natively integrated by Silver Hive.

3. Cab Overview

3.1 Behind the Driver — Battery

Directly behind the driver is the **main battery lever (1)**, which must be engaged before any other operation.

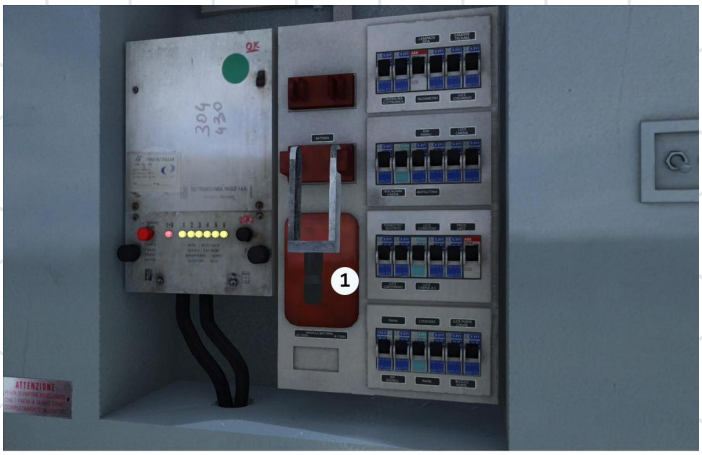


Figure 1 — Behind the driver: battery lever (1)

3.2 Driver's Desk



Figure 2 — Driver's desk with numbered controls

No.	Component	Function
2	Pantographs T1 / T2	Raise and lower pantographs
3	Reverser	Direction: Forward / Neutral / Reverse
4	Notch selector	Traction combinations: 0 M S 0 SP 0 P 0 PP
5	Light selector	Front or rear lights
6	Instrument lighting	Dashboard illumination
11	Wheelslip indicators	Wheel adhesion loss

3.3 Lights, Instruments and Headlights

The light panel houses the **INTERA / RIDOTTA / FANALI** selector for headlight intensity, the **LUCE** button for general cab lighting, and the **LUCE CABINA** and **LUCE STRUMENTI** rotary switches. The indicator lamps (IR, REC, VMf, VM, FAV, IVR, VR, FARO) show the status of on-board electrical systems.

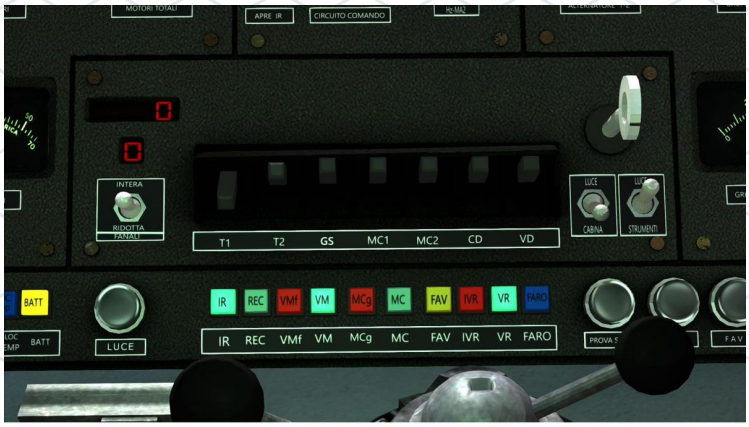


Figure 3 — Light, instrument and headlight panel



Figure 4 — Shunt and dynamic rheostat LEDs

On the panel two seven-segment displays show the traction status: the **upper** display shows the **rheostat** (dynamic), the **lower** one the **shunt**.

3.4 Braking System

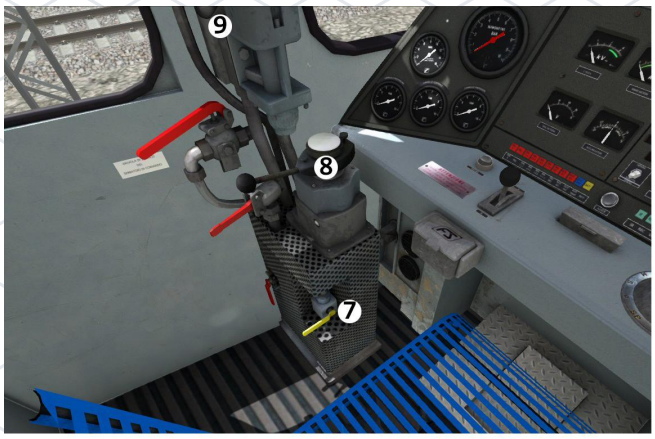


Figure 5 — Isolating cock (7), train brake (8), loco brake (9)

No.	Component	Function
7	Brake isolating cock	Isolates or opens the continuous brake circuit
8	Train brake (Westinghouse)	Brakes the entire train
9	Loco brake (graduated)	Brakes the locomotive only, fully modulable

3.5 Spring Brake

The **spring brake handwheel (10)** behind the driver is the parking brake. Release it fully before any movement; apply only with the locomotive stationary.

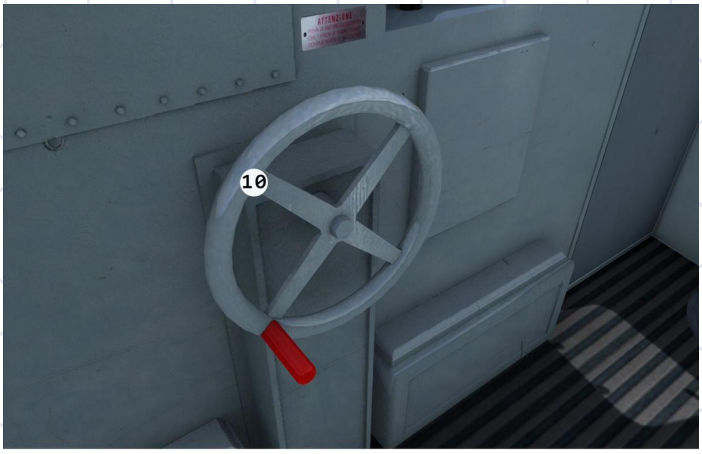


Figure 6 — Spring brake handwheel (10)

3.6 Cabs A and B

The locomotive has **two independent cabs**, one per half-body, with the same controls. They are recognised by the label on the wall: **CABINA ANTERIORE** (cab A, front) and **CABINA POSTERIORE** (cab B, rear).



Figure 7 — Cab A: label “CABINA ANTERIORE”



Figure 8 — Cab B: label “CABINA POSTERIORE”

4. Preparation for Service

Follow the steps in the order shown.

4.1 Battery

1	**Engage the battery lever (1) — [B].** Wait ~10 seconds: the BAT indicator flashes then stays on steady.

> ■ **Warning**
> Do not continue until the BAT indicator is steady. The on-board system needs approximately 10 seconds to initialise.

4.2 Pantographs

2	**Raise pantograph T2 (rear) — [SHIFT+T].** T2 is preferred when travelling forward.
3	**Raise pantograph T1 (front) if needed — [SHIFT+G].** T1 is preferred when travelling in reverse. One pantograph is sufficient for normal driving.

4.3 Spring Brake and Cocks

4	**Release the spring brake (10) — [Backspace].** Turn the handwheel anti-clockwise until fully released.
5	**Open the brake isolating cock (7) — [SHIFT+M].** The continuous brake circuit must be active.

4.4 Starting Traction

6	**Select direction with the reverser (3): Forward [W] or Reverse [S].**
7	**Move the notch selector (4) to position S: [A] up, [D] down.** Increase gradually.
8	**Release the train brake (8): ['] releases, [;] applies.** Reduce pressure progressively.

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5. Driving

5.1 Traction — Notch Selector

Pos.	Name	Use
M	Shunting	Low speeds, yard movements
S	Series	Starting from rest
SP	Series-Parallel	Medium speed
P	Parallel	Cruising speed
PP	Super-Parallel	High speed

5.2 Braking

To decelerate return the notch to 0 ([D]) and apply the **train brake (8)**: [;] applies / ['] releases — acts on the whole train. The **loco brake (9)**: []] applies / [[] releases — loco only. The **spring brake (10)** [Backspace] must only be used with the loco stationary.

5.3 Lights

Key	Function
[H]	Headlights
[SHIFT+F]	Instrument lighting
[E]	Cab lighting
[SHIFT+E]	Driver reading light
[CTRL+E]	Guard's light
[CTRL+T]	Doors
[V]	Windscreen wipers

5.4 Changing cab (A ↔ B)

The cab where the battery lever is operated becomes the **master** cab: the locomotive is driven from there. The other cab is the **slave**. To pass control to the other cab:

1. **Bring the locomotive to a complete stop and put the controls at rest.**
   Speed 0, brake applied, reverser (3) in Neutral and notch selector (4) at 0.
2. **Switch off the battery in the master cab — battery lever (1).**
   Better to operate the lever manually: the [B] key works, but the lever avoids risks.
3. **Move to cab A or B.**
4. **Switch on the battery in the new cab — battery lever (1).**
   Preferably manually. The new cab becomes master.

> ■ **Warning**
> Change cab only with the locomotive perfectly stationary and the master controls at rest.

■ Cab visit
Entering the other cab without switching off the battery in the master is only a visit: control stays with the master.

6. RSC/SCMT System — Silver Hive

6.1 Introduction

The **Silver Hive RSC/SCMT system** reproduces the mandatory safety device on Italian lines equipped with continuous coded track circuits (BAcc). It continuously receives frequency codes from the rails and intervenes if the driver fails to observe signals.

6.2 Activation

Activation requires **two sequential steps** using key [N]: first RSC, then SCMT.

9 **First press [N] — RSC active.**
The RSC blue LED lights steady: the RSC device is active and receiving codes from the line.



Figure 9 — RSC only active (RSC blue LED steady)

10 **Second press [N] — SCMT active.**
The SCMT blue LED lights steady: RAP and overspeed control are now active.



Figure 10 — RSC + SCMT active (both blue LEDs steady)

■ Mandatory order
RSC must be activated before SCMT. If RSC is off, SCMT cannot be enabled.

6.3 RSC Codes

RSC codes are transmitted continuously from the track and shown on the cab display:

Code	Icon	Meaning
270	■ green	Clear — no restriction
180	■ white	Advance warning — acoustic alert only, no action required
120	RV yellow	Immediate diverge — PRE required
75	▲ yel/blk	Next signal at danger — reduce speed
AC	AC white	No code — non-coded section, points or occupied track

6.4 Safety Functions

PRE — Pre-acknowledgement

When code **120** is received the **PRE LED flashes**. Press [X] within **5 seconds**. If not pressed → automatic emergency brake.



Figure 11 — PRE LED flashing: press [X] within 5 seconds

11 **Press PRE as soon as the LED flashes — [X].**
The PRE LED goes out. The RIC LED then lights steady.

■ Warning
If PRE is not pressed within 5 seconds → automatic emergency brake.

RIC — Acknowledgement

After PRE, at the next restrictive code transition the **RIC LED lights steady**. Press [Z]: the system allows **12 seconds** from the press to pass the signal. If not pressed within 12 seconds → automatic emergency brake.



Figure 12 — RIC LED steady: press [Z] then pass the signal within 12 seconds

12 **Press RIC — [Z].**
After pressing, 12 seconds are available to pass the signal.

■ RSC sequence summary
Code 120 → PRE LED flashes → press [X] within 5s → RIC LED steady → press [Z] → 12s to pass signal → system reset.

SR — Signal Passed at Danger

Allows authorised passing of a signal at danger. Must be activated **with the locomotive stationary** in front of the signal. The **SR LED lights red** and remains active for 60 s / max 200 m.



Figure 13 — SR red LED: passing authorised

13 **Stop in front of the red signal, then press SR — [M].**
Proceed past the signal at reduced speed (max 30 km/h).

■ Warning
Passing a red signal without SR active → automatic emergency brake.

RF — Brake Reset

In the event of automatic emergency braking, reset occurs in **two phases**:

Phase	RF LED	Condition	Action
1	Flashing	Train moving	Wait for complete stop — reset not possible
2	Steady	Train stopped	Press RF — [Q] — to release brakes and resume

> ■ **Warning**
> Until RF is pressed, traction remains locked.

RAP — Departure Acknowledgement

Vigilance device active only with **SCMT on**. Each time the loco exceeds 2 km/h a 30-second cycle starts. At the end, an acoustic warning requires pressing [R] within **4 seconds**. If not pressed → emergency brake via RF. The cycle resets automatically at every stop below 2 km/h.



Figure 14 — RAP button on the desk

14 **Press RAP within 4 seconds of the acoustic warning — [R].**
The cycle resets and restarts.

Overspeed Control

If speed exceeds the line limit by more than 2 km/h a double beep sounds every 5 seconds. **No automatic brake is triggered**: this is an acoustic warning only.

6.5 Shunting Mode (MAN)

Press MAN with the loco stationary (< 5 km/h). RSC, SCMT, RAP and overspeed are disabled. Maximum speed **30 km/h**. To deactivate, press MAN again with the loco stationary.



Figure 15 — MAN active: white MAN LED

> **Note**
> If MAN is pressed while the loco is moving the LED does not light and the mode does not activate.

6.6 RSC/SCMT LEDs and Indicators

LED	Colour	State	Meaning
RSC	Blue	Steady	RSC active
SCMT	Blue	Steady	SCMT active (RAP + Overspeed)
MAN	White	Steady	Shunting mode active
PRE	Yellow	Flashing	Press [X] within 5s
RIC	Yellow	Steady	Press [Z] within 12s of press
SR	Red	Steady	SPaD authorisation active
RF	White	Flashing	Emergency brake — train moving
RF	White	Steady	Train stopped — press [Q] to resume
Red tach. LED	Red	Flashing	Automatic emergency brake active

7. Keys and Controls

Key	Function	Key	Function
W	Reverser: Forward	U	Sanders
S	Reverser: Reverse	H	Headlights
A	Notch selector: up	SHIFT+F	Instrument lighting
D	Notch selector: down	E	Cab lighting
SHIFT+T	Pantograph T2 (rear) Up/Down	SHIFT+E	Driver light
SHIFT+G	Pantograph T1 (front) Up/Down	CTRL+E	Guard light
;	Train brake — apply	CTRL+T	Doors
'	Train brake — release	V	Wipers
]	Loco brake — apply	N	RSC/SCMT key
[	Loco brake — release	X	PRE — pre-acknowledgement
Backspace	Spring brake — apply/release	Z	RIC — acknowledgement
SHIFT+M	Brake isolating cock	Q	RF — brake reset
B	Battery On/Off	M	SR — signal passed at danger
Space	Horn	R	RAP — departure acknowledgement

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8. Shutdown

1	**Notch selector to 0 — [D].**
2	**Reverser to Neutral.**
3	**Apply spring brake — [Backspace].**
4	**Isolate train brake — [SHIFT+M].**
5	**Lower pantographs — [SHIFT+T] and [SHIFT+G].**
6	**Deactivate SCMT then RSC — [N] twice.**
7	**Battery Off — [B].**

9. Troubleshooting

Symptom	Solution
BAT indicator flashing for a long time	Wait ~10s after engaging battery [B]
Pantographs will not raise	Check BAT steady, then [SHIFT+T] for T2 or [SHIFT+G] for T1
Traction locked	Check spring brake [Backspace] and RF LED status
RF LED flashing	Wait for train to stop completely
RF LED steady	Train stopped — press [Q] to resume
SCMT will not activate	RSC must be active first: press [N] once for RSC, again for SCMT
PRE unresponsive	Check SCMT is active (blue SCMT LED steady)
MAN will not activate	Locomotive must be stationary (< 5 km/h)
Reporting a problem	Attach the diagnostic report **RailWorks\Log\SH_E656_Classic_Report.txt** (it keeps the last 5 sessions)

10. Credits and Licence

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> ■ **Support**
> For bug reports and updates refer to the official Silver Hive channels.